SMART MOTION LIGHT WITH BATTERY PROTECTION

1. Abstract

The Smart Motion Light with Battery Protection is an embedded system project designed to automate lighting based on human movement. The system uses an ESP32 microcontroller and a PIR motion sensor to detect movement and control the lighting load through a relay module. The project also includes battery-related protection logic in the program. However, battery hardware was not integrated into the final prototype due to a battery fault.

2. Introduction

Energy conservation is an important requirement in modern electrical systems. Lights are often operated unnecessarily when people are not present. This project provides an automatic lighting solution using motion detection and an embedded controller.

3. Objectives

- To develop an automatic lighting control system.
- To detect human movement using a PIR sensor.
- To control lighting using an ESP32 and relay module.
- To reduce unnecessary energy consumption.
- To incorporate battery-related protection logic in the software.

4. Hardware Requirements

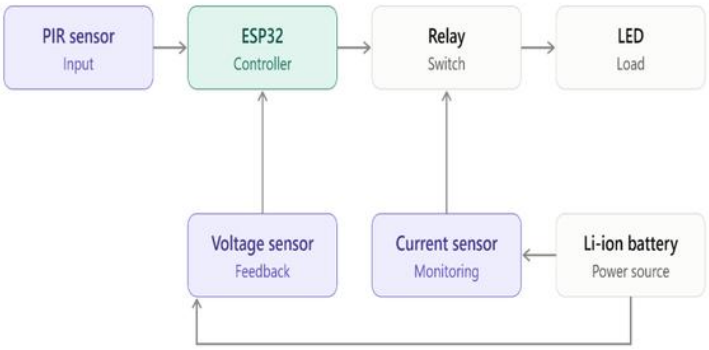
- ESP32 development board
- PIR motion sensor
- Lithium-ion (Li-ion) Rechargeable Batter
- Relay module
- LEDs / lighting indicators
- Breadboard
- Connecting wires

- Power supply

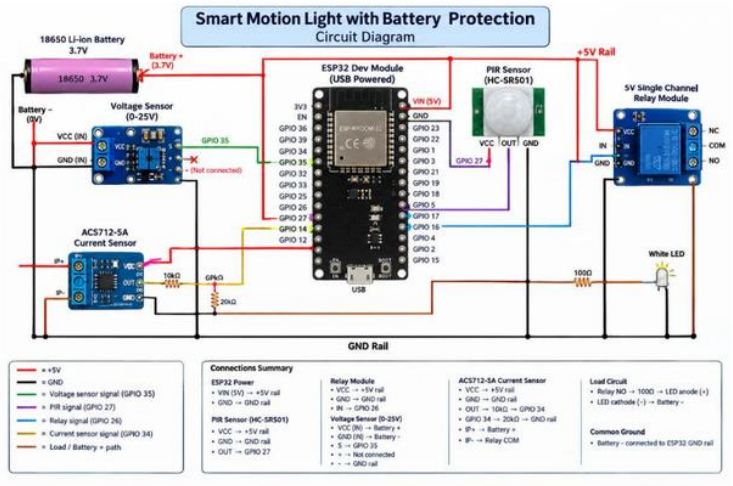
5. Software Requirements

- Arduino IDE
- ESP32 board support package
- Arduino programming language

6. Block Diagram



7. Circuit Diagram



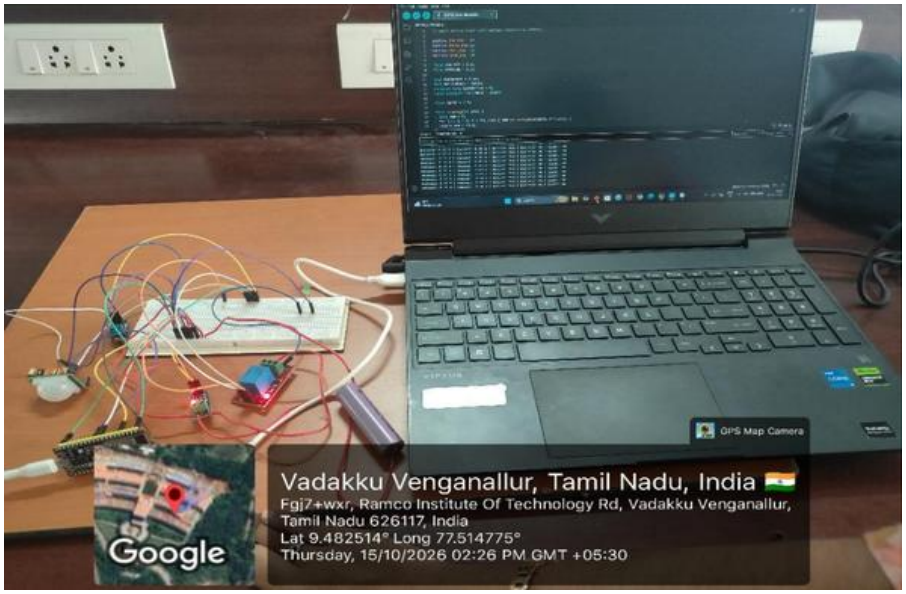
8. Working Principle

The PIR sensor detects human movement and sends a digital signal to the ESP32. The controller processes the input and activates the relay module according to the programmed conditions. When the specified conditions are satisfied, the lighting load is switched ON. When no movement is detected, the controller operates the lighting according to the programmed delay and switching logic.

Battery-related protection logic is included in the software, but battery hardware was not connected in the final prototype.

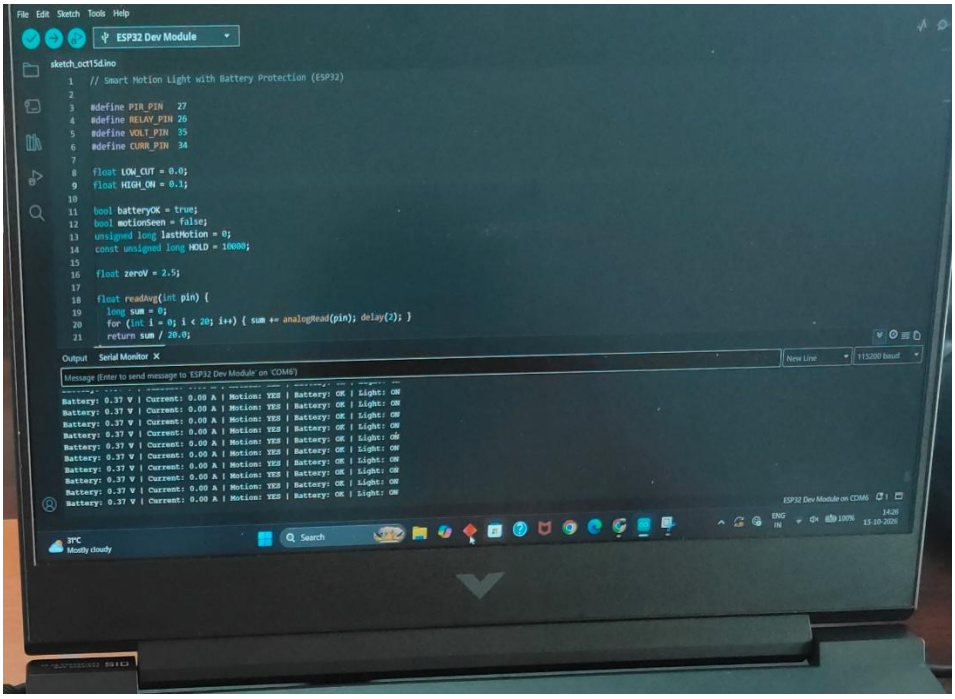
9. Hardware Implementation

The prototype was developed using an ESP32, PIR sensor, relay module and supporting electronic components. The components were interconnected using a breadboard and jumper wires.



10. Results

The hardware prototype demonstrates motion-based lighting control using an ESP32 controller and PIR sensor.



11. Applications

- Smart home lighting
- Corridor lighting
- Staircase lighting
- Energy-efficient lighting systems
- Automatic room lighting

12. Future Enhancements

- Actual battery backup integration
- Battery voltage monitoring
- IoT-based monitoring
- Mobile application control
- Improved power management

13. Conclusion

The project demonstrates an automatic lighting system using motion detection and ESP32-based control. It provides a practical approach to reducing unnecessary lighting operation. Battery protection functionality can be further validated through hardware integration.

14. References

- ESP32 Technical Documentation
- PIR Sensor Datasheet
- Arduino IDE Documentation
- Relay Module Datasheet